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PhD(c) | MSc | Electronic Engineer | Software Developer

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📍 Manizales Caldas - Colombia



Multidisciplinary systems engineer bridging rigorous academic research with advanced software development. With a strong foundation in electronic systems and industrial automation, I specialize in translating complex requirements into highly structured, **deterministic end-to-end solutions**. I bring an engineering-first mindset to the entire product lifecycle, seamlessly connecting robust backend architectures with intuitive user interfaces. Thriving in high-autonomy environments, I focus on deep analytical problem-solving to ensure that innovative concepts systematically evolve into reliable, **production-ready software**.

SKILLS

Architecture & System Design	Microservices, Distributed Systems, Event-Driven Architecture, Observability, Deterministic Execution
Programming Languages	Python, TypeScript, JavaScript, Bash, HTML5, CSS3, SQL
Backend Ecosystem	FastAPI, Django (DRF, Channels), Flask, asyncio, Celery
Frontend Ecosystem	Angular (RxJS, Signals), React, Next.js, Tailwind CSS, SASS
Artificial Intelligence	AI Agent Architectures, RAG Systems, LLMops, Prompt Engineering, PyTorch, TensorFlow
Databases & Vector Stores	PostgreSQL, TimescaleDB, pgvector, ChromaDB, Redis, SQLite
DevOps & Cloud	GCP, AWS, Docker, Kubernetes, Docker Swarm, GitHub Actions, CI/CD
Observability & Tools	Git, Prometheus, Grafana, Apache Kafka, Nginx
Operating Systems	Linux Administration (Arch Linux, Ubuntu Server), MacOS

TL;DR

Senior **Full-Stack Engineer** with 10+ years of experience in *Python* and a strong track record building **scalable backend architectures** and distributed systems. I specialize in orchestrating *microservices*, managing concurrent workloads with **asyncio**, and designing production-ready systems using *FastAPI* and *Django*. My expertise bridges advanced backend development, containerized infrastructure via **Kubernetes**, and data-driven applications, ensuring reliability and traceability through comprehensive **observability stacks** like Prometheus and Grafana.

I have extensive experience architecting data pipelines and **real-time streaming** workflows, particularly within high-throughput and research environments. This includes a deep focus on *Artificial Intelligence*, where I design and deploy deterministic **AI agent architectures**, complex **RAG systems**, and tailored **LLM integrations**. My work ensures that advanced machine learning models are seamlessly connected to robust backend services and structured data layers, maintaining high performance and predictability.

In parallel, I drive full-stack delivery by connecting highly optimized backend systems with modern, interactive user interfaces built on *TypeScript* and **Angular**. From **infrastructure-as-code** and automated **CI/CD pipelines** to front-end state management, I focus on engineering clean, maintainable codebases that support the full lifecycle of complex, cloud-native products.

Core Engineering Expertise

- **Backend & Concurrency:** Advanced Python async architectures via **asyncio**, FastAPI, and scalable Django systems.
- **Data Systems:** Complex SQL schema design, PostgreSQL optimization, and high-throughput TimescaleDB time-series streaming.
- **AI & Agents Engineering:** Production-ready RAG architectures, LLM orchestration, and deterministic AI agent workflows.
- **Observability:** Zero-to-production monitoring pipelines using Prometheus, Grafana, and structured telemetry/logging.
- **Security & Identity:** Production-grade implementations of OAuth2, JWT, and cryptographically secure service-to-service communication.

- **Container Orchestration:** Cloud-native deployment pipelines utilizing Docker, Docker Swarm, and production Kubernetes clusters.
- **CI/CD & GitOps:** Automated testing with PyTest, coverage enforcement via pre-commit hooks, and complex GitHub Actions workflows.
- **Systems Automation:** Advanced Bash scripting, Makefiles, Infrastructure-as-Code (Ansible), and custom CLI operational tools.

Highlighted Projects

- **Deterministic AI Frameworks:** Production runtime bridges and agent-driven architectures engineered to eliminate LLM unpredictability.
- **ICU Monitoring System:** High-throughput, real-time biosignal ingestion platform featuring automated alert thresholds and metrics dashboards.
- **Chaski-Confluent Pipeline:** Distributed message-driven pipeline integrating TensorFlow analytics and deterministic execution scheduling.
- **BCI Research Infrastructure:** Specialized EEG data acqui-

Infrastructure & Automation

sition and processing framework with end-to-end built-in system observability.

› **CtenoLab Control Suite:** Web-based experimental control

ecosystem managing stimulus delivery and low-latency feedback loops.

PROFESSIONAL EXPERIENCE

From November 2025
To Present

Self-employed

LEAD SOFTWARE ARCHITECT & RESEARCHER
DunderLab | Manizales, Colombia

Dedicated to the research and development of internal tools and proprietary software solutions, focusing on authoring and submitting advanced research proposals within my doctoral field. This period involves the design and implementation of internal MLOps workflows to optimize Large Language Model (LLM) deployments, alongside the orchestration of scalable infrastructure for automated data processing and RAG-based systems.

Python FastAPI MLOps LLMs RAG Infrastructure

From September 2025
To November 2025

Full-time

SOFTWARE ENGINEER (RESEARCH PROJECT)
Universidad Nacional de Colombia | Manizales, Caldas

Designed and implemented the full software stack of a remote RF-sensing platform deployed on a secure VPS environment. Developed backend services and WebSocket interfaces for real-time interaction with distributed SDR-based sensors; implemented spectral-analysis capabilities including FFT processing (RBW/VBW control), noise-floor estimation, occupied-bandwidth metrics, peak/marker tools, and automatic detection of unauthorized emissions. Built device-management, user-access, monitoring, and measurement-scheduling modules as described in the official platform manual. Ensured secure data transport (TLS), OAuth-based authentication, and integration with geolocation and logging services, enabling reliable nationwide spectrum-monitoring operations.

Strategic Planning Research Consulting Developer

From May 2024
To August 2025

Hourly-based

ADJUNCT PROFESSOR
Universidad Nacional de Colombia | Manizales, Caldas

Conducted in-person teaching of the undergraduate course "Digital Signal Processing" for the Electronic Engineering program at the Faculty of Engineering and Architecture, Universidad Nacional de Colombia. Responsibilities included preparing lectures, guiding practical sessions, and evaluating student performance.

Adjunct Professor

From February 2025
To April 2025

Full-time

TECHNICAL LEAD
DISERED | Bogotá

Led and coordinated a multidisciplinary team of software engineers in the design, development, and deployment of scalable microservices architectures. Defined technical strategies, ensured adherence to software engineering best practices, and maintained high standards of code quality, system reliability, and performance. Translated business requirements into robust technical solutions, guided architectural decisions, oversaw CI/CD pipelines, and fostered collaboration between technical and non-technical stakeholders. Also mentored team members and promoted innovation through modern technologies and agile methodologies.

Python FastAPI Docker PostgreSQL Google Cloud Platform

From May 2024
To December 2024

Full-time

SOFTWARE ENGINEER (RESEARCH PROJECT)
Universidad Nacional de Colombia | Manizales, Caldas

Provided professional services as an electronic engineer with a master's degree in Industrial Automation to support the conceptual design, requirement specification, implementation, and configuration of software-defined radio (SDR) sensors. Delivered a cost-efficient and scalable prototype for radio spectrum monitoring in Colombia, integrating SDR technology with deep learning techniques.

Python SDR Deep Learning Prototyping

From June 2023 To September 2024 Full-time	RESEARCH ENGINEER (PROJECT LEAD) Universidad Nacional de Colombia Manizales, Caldas Led research and development of a functional prototype of an electronic tongue aimed at identifying flavor profiles in fine cocoa of Colombian origin. Oversaw sensor integration, designed signal processing workflows, and directed system validation. Signal Processing Sensor Integration Prototype Validation
From June 2023 To March 2024 Full-time	RESEARCH CONSULTANT (STRATEGIC PLANNING) Universidad Nacional de Colombia Bogotá Designed and proposed a strategic plan for the establishment of the inter-campus Center of Excellence in Medicine and Artificial Intelligence (SEMAI), as part of the national call for the consolidation of centers of excellence 2020–2021. Strategic Planning Research Consulting AI in Medicine
From January 2023 To December 2023 Full-time	SOFTWARE ENGINEER (PROJECT LEAD) DunderLab/Universidad Tecnológica de Pereira Pereira, Risaralda Led the design and implementation of a gait pattern monitoring system, focusing on data acquisition, processing, and visualization to support movement analysis and rehabilitation applications. Oversaw technical decisions and coordinated development tasks to ensure system reliability and applicability. Python Data Acquisition Signal Processing Visualization
From January 2023 To May 24, 2023 Full-time	PROJECT LEAD — INFORMATION MANAGEMENT TOOL DunderLab/Universidad Tecnológica de Pereira Pereira, Risaralda Directed the design and development of an information management tool to support the planning and coordination of Colombia's national natural gas system. Oversaw backend design, data modeling, and system integration, ensuring scalability and reliability of the solution. Python PostgreSQL Data Modeling System Integration
From March 2023 To May 2023 Full-time	ENGINEERING CONSULTANT Universidad Nacional de Colombia Manizales, Caldas Designed and led the development of a protocol for a biosignal acquisition and capture system to monitor cardio-cerebral-pulmonary activity in Neonatal Intensive Care Unit (NICU) patients. Directed the creation of a prototype applying machine learning techniques to biosignal processing, with initial validation in a simulated environment. ESP32 MicroPython PostgreSQL Django Django Rest Framework
From September 2022 To December 2022 Full-time	ENGINEERING CONSULTANT Universidad Nacional de Colombia Manizales, Caldas Spearheaded the development of an information visualization system for campus-level research, extension, and innovation indicators. Integrated the system with the Research and Extension Department's website, supported outreach activities with local and national entities, and contributed to the office's strategic planning structure. Python Django Dash PostgreSQL JavaScript HTML CSS
From September 2022 To December 2022 Full-time	SOFTWARE ENGINEER (ULTRASOUND LOCALIZATION) DunderLab/Universidad Tecnológica de Pereira Pereira, Risaralda Led the development of backend and user interface modules for a software tool focused on reconstruction and localization, as part of the project "Development of a needle tracking and nerve localization tool in ultrasound for regional anesthesia practice." Directed integration efforts to support treatment of acute traumatic pain and prevention of chronic neuropathic pain. Python Django TensorFlow Matplotlib Scikit-learn
From June 2022 To September 2022	SOFTWARE ENGINEER (MEDICAL THERMOGRAPHY) DunderLab/Universidad Nacional de Colombia Manizales, Caldas

Full-time	Developed software for thermographic measurement and regional anesthesia monitoring, as part of the project “Support tool for predicting the effects of local anesthetics via neuroaxial epidural using infrared thermography.” Delivered a solution integrating image analysis techniques for clinical decision support. Python PySide Raspberry Pi GNU/Linux
From January 2022 To March 2022	RESEARCH ENGINEER (EEG PREPROCESSING) DunderLab/Universidad Tecnológica de Pereira Pereira, Risaralda
Full-time	Designed and fine-tuned preprocessing algorithms for EEG recordings, enhancing the acquisition module of a prototype system for neuropathic pain diagnosis. Improved signal quality for accurate downstream analysis. Python Kafka Brython Tornado HDF5 OpenBCI
From December 2021 To January 2022	SOFTWARE ENGINEER (NEUROPHYSIOLOGICAL SIGNALS) DunderLab/Universidad Nacional de Colombia Manizales, Caldas
Full-time	Designed and implemented a neurophysiological signal acquisition module within the research project “Enhanced and interpretable deep learning framework to support computer-assisted diagnosis systems.” Enabled real-time biosignal capture for machine learning applications in healthcare. Python Kafka Brython Tornado
From July 2021 To November 2021	SOFTWARE ENGINEER (ADHD DIAGNOSTIC UI) DunderLab/Universidad Tecnológica de Pereira Pereira, Risaralda
Full-time	Delivered a web-based user interface for a diagnostic support tool for Attention Deficit Hyperactivity Disorder (ADHD), enabling intuitive access to data visualization and clinical resources. Python PySide Kafka HTML CSS Brython
From September 2019 To May 2021	RESEARCH ENGINEER (MEDICAL IMAGING - NEUROSTIMULATION) Universidad Nacional de Colombia Manizales, Caldas
Full-time	Led morphological characterization of brain structures using medical imaging to support surgical implantation of neurostimulators for Parkinson’s disease treatment. Developed image classification methodologies, registration algorithms, and contributed to the construction of anatomical atlases. Python PySide Kafka HTML CSS Brython OpenBCI Raspberry Pi JavaScript Django Numpy Scipy Matplotlib

PROJECTS

Throughout my career, I have led and developed individual and collaborative projects in software frameworks, signal processing, and web infrastructure. These works highlight my focus on building open, modular, and scalable solutions—from real-time neuroscience platforms to developer frameworks and backend systems. Below is a selection of representative open-source projects I have designed, maintained, or released.

CTENOLAB

FROM 2025 - TO PRESENT

 cteno.dunderlab.com

Designed and developed a real-time cognitive experimentation platform integrating neurophysiological paradigms with hardware triggers over TCP/IP. Implemented distributed communication with dynamic node discovery, subscription-based messaging, and real-time routing. Added remote Python execution, TensorFlow Serving integration, and AI-driven broadcasting for experimental control.

Python TCP/IP SWARM API Docker TensorFlow RPyC WebUSB WebSerial

REMOTE SPECTRAL SENSING PLATFORM

FROM 2024 - TO 2025

cteno.dunderlab.com

Developed a full remote RF-sensing platform enabling real-time spectrum acquisition, processing, and device orchestration over distributed SDR nodes. Implemented FFT-based spectral analysis (RBW/VBW control, peak/marker tools, OBW/-XdB bandwidth metrics), automated detection of unauthorized emissions, and geolocated monitoring from multiple remote sensors. Built backend WebSocket services for low-latency data streaming, a full device-management and scheduling system, and secure TLS/OAuth access. Deployed the entire stack on a hardened Linux VPS with continuous monitoring and persistent storage of spectral datasets.

FFT/Signal Processing SDR (HackRF) TLS OAuth PostgreSQL/MySQL Linux VPS Angular Django REST Framework

RADIANT WRAPPER

FROM 2023 - TO 2025

github.com/dunderlab/radiant-wrapper

Developed a Python-to-Android build system that automates the packaging of static HTML and Django projects into Android APKs. Implemented pre-built Docker images with NDK/SDK integration and automated GitHub workflows for reproducible builds. Added support for environment detection, CI/CD pipelines, and lightweight deployment of research and MVP applications.

Python Docker GitHub Actions Android NDK Django HTML

CHASKI-CONFLUENT

FROM 2022 - TO PRESENT

chaski.dunderlab.com/ github.com/dunderlab/python-chaski

Designed and developed a distributed communication framework over TCP/IP featuring dynamic node discovery, subscription-based messaging, and real-time message routing. The system includes support for remote Python execution, TensorFlow Serving integration, and AI-driven message broadcasting.

Python TCP/IP SWARM API Docker Tensorflow RPyC

FOUNDATION

FROM 2022 - TO PRESENT

github.com/dunderlab/python-dunderlab.foundation

Created a modular Python framework to build scalable and distributed systems. Provides utilities for data pipelines, inter-process communication, and infrastructure orchestration with an emphasis on flexibility and system decoupling.

Python Docker Distributed Systems Modular Architecture

DOCS: AUTOMATED SPHINX DOCUMENTATION GENERATION

2023

github.com/dunderlab/python-dunderlab.docs dunderlab-docs.readthedocs.io

Built a Python module to automate the generation of Sphinx documentation from Jupyter Notebooks using nbsphinx. Includes features such as automated README creation, HTML index generation, and seamless Docker and GitHub workflows integration.

Python Sphinx nbsphinx Jupyter Notebooks Docker GitHub Workflows

BCI-FRAMEWORK

FROM 2018 - TO 2022

docs.bciframework.org github.com/UN-GCPDS/bci-framework

Developed a modular framework for designing and running psychophysiological experiments with OpenBCI. Included real-time data visualization, distributed stimulus delivery, and signal processing tools for research workflows.

Python Tornado Matplotlib Brython Kafka OpenBCI

OPENBCI-STREAM

FROM 2018 - TO 2022

openbci-stream.readthedocs.io github.com/UN-GCPDS/openbci-stream

Created a high-level module for real-time EEG, EMG, and ECG data acquisition and streaming using the OpenBCI Cyton board, enabling distributed neuroscience experiments and analysis.

Python Kafka OpenBCI

MATPLOTLIB-FIGURESTREAM

FROM 2019 - TO 2022

figurestream.readthedocs.io github.com/UN-GCPDS/matplotlib-figurestream

Built a lightweight backend to stream Matplotlib animations over the web, allowing real-time visualizations in distributed systems and remote monitoring tools.

Python Matplotlib Flask

QT-MATERIAL

FROM 2019 - TO 2022

qt-material.readthedocs.io github.com/UN-GCPDS/qt-material

Designed and maintained a Material Design-inspired stylesheet for PyQt and PySide applications. Focused on visual consistency and improved UX across platforms.

Python PySide PyQt6 CSS QT

RADIANT RUNTIME BRIDGE

FROM 2019 - TO 2022

github.com/dunderlab/radiant-runtime-bridge

Engineered a Brython-based framework for building responsive web apps entirely in Python. Integrated frontend logic with Tornado for dynamic, reactive interfaces.

Python Tornado Brython

PINGUINO PROJECT

FROM 2012 - To 2015

pinguino.cc github.com/PinguinoIDE

Contributed to the development of an open-source electronics prototyping platform and IDE, supporting Microchip micro-controllers and aiming to simplify embedded system design.

Python PySide Microchip

 LANGUAGES

English ●●●○○
Spanish ●●●●●

 STRENGTHS

- > Enthusiast
- > Motivated
- > Autonomous

 FORMATION

currently	PhD in Automation Engineering, Universidad Nacional de Colombia
2022	MSc in Industrial Automation, Universidad Nacional de Colombia
2018	BSc in Electronic Engineering, Universidad Nacional de Colombia